

MUHAMMAD UMAR ALI

umar.ali@uwaterloo.ca | +1 (778) - 779 - 2078 | [linkedin.com/in/muhammad-umar-ali](https://www.linkedin.com/in/muhammad-umar-ali) | [umarali.ca](https://www.umarali.ca) | github.com/gitUmaru

SKILLS

Languages: Python, Java, C#, C/C++, MATLAB, R, JavaScript, Dart, Bash, HTML/CSS, and $\text{\LaTeX}$
Frameworks & Tools: Git, Docker, Jira, Figma, Flutter, JUnit, Android Studio, and High Performance Computing
Libraries: TensorFlow, Keras, Pytorch, OpenCV, Pandas, NumPy, Scipy/Scikit-learn, Matplotlib, Seaborn

EDUCATION

University of Waterloo Waterloo, ON
Doctor of Philosophy in Systems Design Engineering Sep 2024 – May 2029
• **Awards:** Ontario Graduate Scholarship, University of Waterloo Presidential Scholarship

University of British Columbia Vancouver, BC
Bachelor of Applied Science in Biomedical Engineering - Concentration in Bioinformatics Sep 2019 – May 2024
• **Awards:** Dean's Honour List, NSERC USRA, Graduated with Distinction

EXPERIENCE

University of Waterloo Waterloo, ON
Sessional Lecturer May 2026 – Aug 2026
• Created and delivered lecture material on data structures and algorithms to class of **100+ first year mechatronic students**.
• Designed tutorial and laboratory around solving Leetcode questions to prepare students for real world coding interviews.

University of Waterloo Waterloo, ON
Teaching Assistant Jan 2025 – Nov 2025
• Lead and developed tutorials for systems modeling and dynamics of mechanics for **60+ third-year engineering students**.
• Designed and built a custom automated grading server to evaluate deep learning submissions of **30+ graduate students**.

Aspect Biosystems Vancouver, BC
Applications Software Engineer Jun 2023 – Aug 2024
• Led web development effort for bolstering full-stack therapeutic platform contributing to securing **\$200 million** investment.
• Spearheaded development of desktop application for microfluidic printhead quality control, increasing production by **11%**.

Systems Software Engineer Sep 2022 - Apr 2023
• Independently developed a minimum viable product of a Windows Desktop application for device quality assurance during Novo Nordisk partnership negotiations, playing a role in **\$2.6 billion** partnership negotiations.
• Enhanced STM32 messaging scheme for asynchronous and parallel write operations, reducing run time cost by **68%**.

BC Cancer Research Center Vancouver, BC
Deep Learning Research Assistant May 2021 – Aug 2023
• Trained generative deep learning models to create skin disease images in patients of colour improving image diversity.
• Accepted to present at several conferences (Harvard National Collegiate Research Conference 2024, SBME Symposium 2023, BC Cancer Summit 2021) **receiving commendations** for excellent scientific communication.

Artificial Intelligence in Medicine Laboratory Vancouver, BC
Machine Learning Research Assistant May 2020 – July 2020
• Designed, developed and implemented cross-platform machine learning model using ONNX and TFLite with accuracy of **90%**
• Fine-tuned ResNet50 network for coarse classification on ovarian carcinoma histotypes for **improved slide scanning**.

PROJECTS

WikiServer | Java, ANTLR, JUnit, gradle, jwiki, gson Nov - Dec, 2020
• Implemented thread safe buffer datatype for efficient querying of Wikipedia entries using Java concurrency and generics.
• Parsed and executed several structured queries on memory managed threads with caching to handle large request loads.

EXTRACURRICULAR

Ibtikar Waterloo, ON
Vice President of Research Network Jan 2025 – Present
• Launched monthly research seminar series to highlight underrepresented research across all faculties in Muslim and other minority communities; consistently reaching **20+ attendees**.
• Helped premiere a Socratic dialogue event to foster critical discussion, intellectual engagement, make complex social and academic topics more digestible to the general public; consistently **reaching 30+ attendees**.